SMARTCAMPUS HYBRID AI INTEGRATION

Technical Specification & Implementation Guide — v1.0 · March 2026
Confidential · Internal Documentation

Audience: Development Team · Product Team · School Administrators
Related Modules: Student Portal, Teacher/Admin Dashboard, Backend Services, Analytics, Communication

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1. Overview

The **Hybrid AI Integration** combines **free AI tools** (ChatGPT 3.5, Perplexity Basic) for students with **paid AI services** (GPT-3.5/4, Perplexity Pro API) for teachers and administrators. The system routes users according to their **roles** and provides **secure, role-based access** to AI features while optimizing cost.

Goals:

- Reduce operational costs for student usage.
- Enable advanced automation for administrative workflows (grading, reports, analytics).
- Maintain data privacy and compliance with Kenyan Data Protection regulations.
- Deliver an intuitive, role-aware AI experience.

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2. Key Components

Component	Description
Frontend Portals	Student portal, Teacher/Admin dashboard, AI input interfaces.
SmartCampus Backend	Handles authentication, role-based routing, API calls, and logging.
Free AI Tools	GPT-3.5 (ChatGPT web), Perplexity Basic embedded via widgets or links.
Paid AI Services	GPT-3.5/4 APIs, Perplexity Pro API; used for automation tasks like grading, report summaries, and analytics.
Role-Based Access Control (RBAC)	Ensures users only see modules and AI features permitted by their role.
API Gateway	Secures and manages calls to external AI services.
Audit & Logging	Records API usage, queries, and responses for security and billing analysis.

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3. Workflow Description

3.1 Student Path (Free AI Access)

1. Student logs in → SmartCampus backend identifies role.
2. System routes to **Student AI interface**:
    - Option 1: Direct free AI link (opens ChatGPT 3.5 / Perplexity in a new tab).
    - Option 2: Embedded AI widget (keeps student inside the portal).
3. Queries are sent to **free AI tools**. Responses are rendered in the student interface.
4. No cost is incurred for student usage.

3.2 Teacher/Admin Path (Paid AI Access)

1. Teacher/Admin logs in → SmartCampus backend identifies role.
2. System routes to **Teacher/Admin AI interface** with automation features (grading, analytics).
3. AI query input → backend → **API call** to GPT-3.5/4 or Perplexity Pro.
4. Responses returned to dashboard → can generate reports, summaries, or analytics.
5. Cost is incurred per API usage.

3.3 Backend Responsibilities

- Authenticate users and determine roles.
- Route queries to appropriate AI tools (free vs paid).
- Track API usage and log queries/responses.
- Enforce RBAC and prevent unauthorized access.

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4. Technical Specifications

Feature	Specification
User Authentication	Standard SmartCampus login (username + password). Supports MFA for admin/finance roles.
Role-Based Routing	Backend checks roles → auto-directs to: • Single-module access → auto-land • Multi-module access → dashboard filtered by role
AI Query Handling	• Student → free AI widget or link • Admin/Teacher → backend API call • All calls logged for auditing
API Security	• HTTPS endpoints • API keys encrypted in backend • Rate limiting per role
Data Storage	Query logs and summaries stored in **SmartCampus DB**; student data anonymized for AI services if needed.
Integration	AI modules integrate with: • Grading/Report modules • Analytics module • Communication module for notifications
Scalability	Supports multiple campuses; API calls queue and cache for peak loads.
Compliance	Role-based access ensures minimal exposure; external AI services comply with Kenyan DPA.

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5. Implementation Steps

Phase 1 — Setup

1. Register accounts/API keys with OpenAI and Perplexity.
2. Configure SmartCampus backend to support role-based routing.
3. Embed free AI links/widgets in student portal.

Phase 2 — API Integration

1. Develop backend API gateway to securely send queries to GPT-3.5/4 and Perplexity Pro.
2. Map teacher/admin automation tasks (grading, summaries, analytics) to API endpoints.
3. Implement query logging, error handling, and rate limiting.

Phase 3 — Role-Based UX

1. Filter dashboard menus and modules per role.
2. Auto-route users based on assigned modules and last activity.
3. Test multi-role accounts (e.g., teacher + parent).

Phase 4 — Security & Compliance

1. Enable MFA for sensitive roles.
2. Encrypt API keys and log sensitive data access.
3. Review queries sent to external APIs for student privacy compliance.

Phase 5 — Analytics & Cost Monitoring

1. Track API usage per user and module.
2. Generate monthly cost estimates for admin review.
3. Optimize query batching to reduce token usage.

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6. Example API Flow (Teacher/Admin)

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Teacher/Admin Input -> SmartCampus Backend -> API Gateway -> GPT-3.5/4 or
Perplexity Pro
Response -> Backend -> Admin Dashboard -> Reports/Grades/Analytics
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7. Security & Compliance Checklist

Requirement	Status
HTTPS for all AI requests	☐
API Key encryption	☐
Role-based access enforcement	☐
Audit logging of queries/responses	☐
MFA for admin/finance roles	☐
Student data anonymization	☐
Compliance with Kenyan DPA	☐

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8. Cost Management

- Free AI tools: $0/student → ideal for high-volume student queries.
- Paid AI services: billed per token or subscription → recommended only for automation-heavy admin tasks.
- Hybrid approach ensures cost efficiency while enabling advanced functionality.

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9. Maintenance & Monitoring

1. Monitor API key usage and expiration.
2. Audit logs weekly for unusual activity.
3. Update AI models/APIs when new versions are released.
4. Optimize student queries to reduce free-tier limitations.